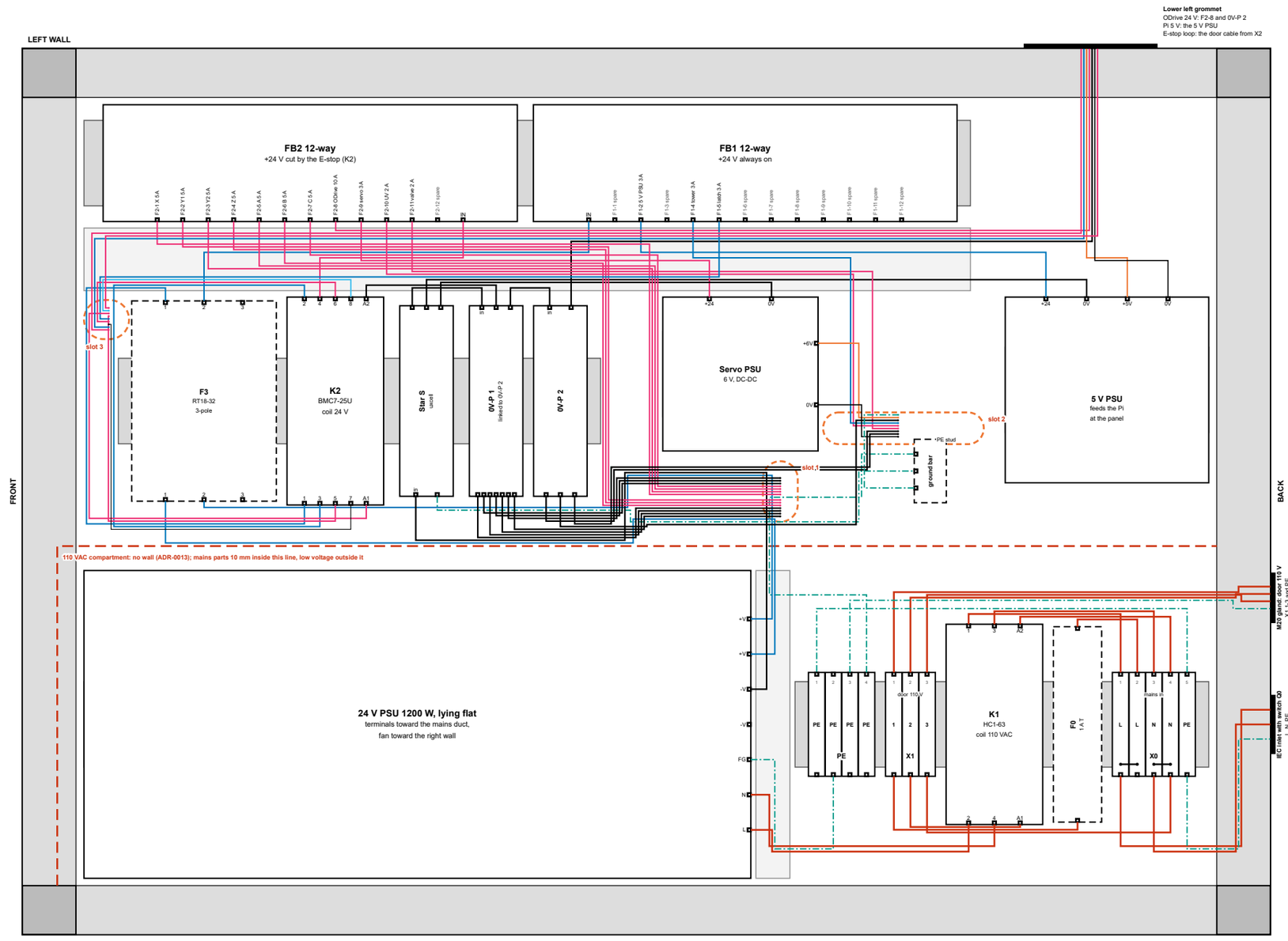
RMR control panel: lower deck wiring

Lower deck, on the base plate: 110 VAC and 24 V power. Sheet 1's circuits on the parts of sheet 3, every wire from terminal to terminal.



Legend

- 110 VAC
- +24 V behind K2 (FB2)
- 0V-S signal 0 V
- +6 V servo, +5 V Pi
- signal
- motor cable
- wire duct
- cable slot (dashed: overhead)
- grommet, inlet or gland
- +24 V always on (FB1)
- 0 V power (0V-P, star S)
- PE
- their 0 V
- multi-core signal cable
- DIN rail
- wire ends in a slot
- part (dashed: size assumed)

Wires between the decks

Slot 1 (x 267.5-280.5, y 156-178)

- F2-1 to driver X +V
- F2-2 to driver Y1 +V
- F2-3 to driver Y2 +V
- F2-4 to driver Z +V
- F2-5 to driver A +V
- F2-6 to driver B +V
- F2-7 to driver C +V
- 0V-P 1 to driver X 0V
- 0V-P 1 to driver Y1 0V
- 0V-P 1 to driver Y2 0V
- 0V-P 1 to driver Z 0V
- 0V-P 1 to driver A 0V
- 0V-P 1 to driver B 0V
- 0V-P 1 to driver C 0V

Slot 2 (x 290-350, y 136-149)

- F2-10 to SSR-UV +
- F2-11 to SSR-valve +
- F1-4 to the tower SSRs + (jumpered)
- 0V-P 2 to X6 UV-
- 0V-P 2 to X6 V-
- 0V-P 2 to X7 C (tower common)
- servo PSU +6 V to X5 6V
- servo PSU 0 V to X5 0V
- lower ground bar to upper ground bar

Slot 3 (x 23-36, y 90-106)

- F1-5 to X2 1 (E-stop latch)
- K2 5 to X2 2b (seal-in, node X)
- K2 6 to X2 3 (node Y)
- K2 A1 to X2 3b (coil, node Y)
- K2 7 to Mega 43 (motor power sense)
- K2 8 to 0V-S
- door cable: X2 1 to S2 NC1, via the lower left grommet
- door cable: NC1 and S3 (node X) to X2 2
- door cable: S3 (node Y) to X2 3

Notes

Seen from above, front at the left, so the left wall is at the top. Each part sits where sheet 3 puts it; ducts and channels are drawn wider than scale so that every wire shows: measure from sheet 3. Terminals are drawn on the face their wires leave from; read the real order off each part.

Since 2026-10-15 the Pi sits at the operator panel on the 5 V PSU, the Mega runs on the PI's USB and the screen has no supply of its own: F1-1, F1-3, X8 and X2 4-5 are spare (NC2 and the home button are wired at the panel). The ODrive sits in the spin coater's base.

Gauges (issue #205): mains, the PSU to F3 and S, F3 to K2 and FB1, K2's pole jumper, K2 to FB2, S to 0V-P and the 0V-P link: 12 AWG copper. F2-1 to F2-8 with their 0 V returns, and the servo 6 V and Pi 5 V pairs: 14 AWG copper. The F0 rung: 18 AWG UL 1015, 600 V. Everything else: the 20 AWG on hand, copper-clad aluminum (about 3 A), only on fuses of 3 A or less.

PE home runs of the frame, the DIN rails and the TB6600 cases go to the ground bars (sheet 4). 0 V meets PE only at star S, by its one wire to the lower ground bar.

Open: tower common polarity, lid feed, valve coil, servo 6 V rating, J11 logic supply, and the 5 V PSU's current for the Pi 5 (5.1 V, 5 A).

Mains duct: a separator between the 110 VAC wires and the PSU's DC and PE leads. Every 110 VAC terminal finger-safe or covered (ADR-0013). Fuses: F3-1 25 A and F3-2 10 A (10 x 38 mm, DC), F2-1 to F2-7 5 A, F2-8 10 A, F2-9 3 A, F2-10 and F2-11 2 A; F1-2, F1-4 and F1-5 3 A; F0 1 A T.

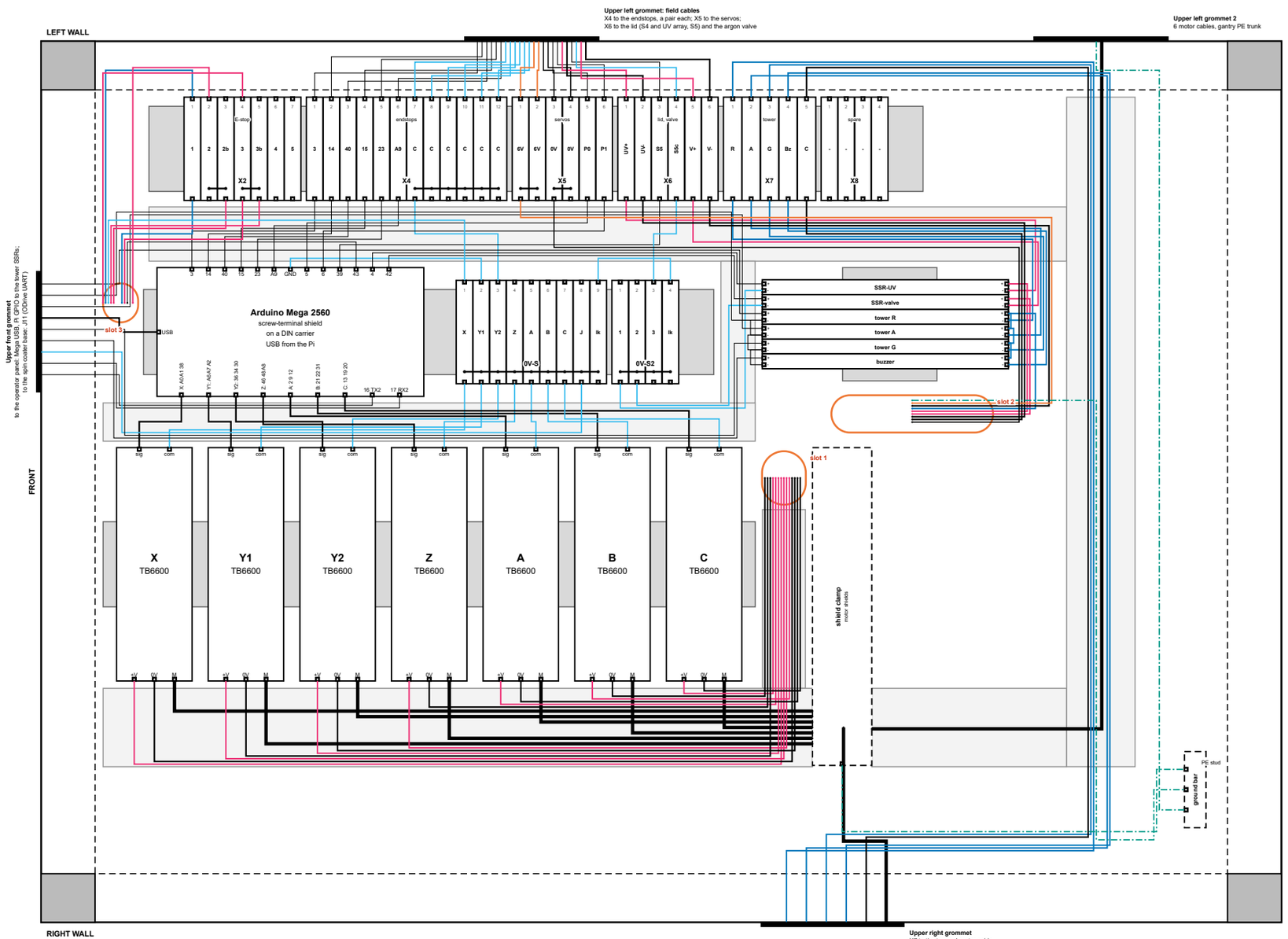
The PSU's terminal order is assumed.

RMR control panel: lower deck wiring

Sheet 2A (2B: upper deck). Legal 14 x 8.5 in, landscape, 100 %.
Robot-MakingRobot, docs/wiring. Drawn 2026-10-19 (issue #164), revised 2026-10-22.
Status: design, not yet built or tested.
Circuits: sheet 1. Positions: sheet 3. Frame and PE: sheet 4.

RMR control panel: upper deck wiring

Upper deck, on the upper plate: drives, logic and field terminals. Sheet 1's circuits on the parts of sheet 3, every wire from terminal to terminal.



Legend

- 110 VAC
- +24 V behind K2 (FB2)
- 0V-S signal 0 V
- +6 V servo, +5 V Pi
- signal
- motor cable
- wire duct
- cable slot (dashed: overhead)
- grommet, inlet or gland
- +24 V always on (FB1)
- 0 V power (0V-P, star S)
- PE
- their 0 V
- multi-core signal cable
- DIN rail
- wire ends in a slot
- part (dashed: size assumed)

Wires between the decks

Slot 1 (x 267.5-280.5, y 156-178)

- F2-1 to driver X +V
- F2-2 to driver Y1 +V
- F2-3 to driver Y2 +V
- F2-4 to driver Z +V
- F2-5 to driver A +V
- F2-6 to driver B +V
- F2-7 to driver C +V
- 0V-P 1 to driver X 0V
- 0V-P 1 to driver Y1 0V
- 0V-P 1 to driver Y2 0V
- 0V-P 1 to driver Z 0V
- 0V-P 1 to driver A 0V
- 0V-P 1 to driver B 0V
- 0V-P 1 to driver C 0V

Slot 2 (x 290-350, y 136-149)

- F2-10 to SSR-UV +
- F2-11 to SSR-valve +
- F1-4 to the tower SSRs + (jumpered)
- 0V-P 2 to X6 UV-
- 0V-P 2 to X6 V-
- 0V-P 2 to X7 C (tower common)
- servo PSU +6 V to X5 6V
- servo PSU 0 V to X5 0V
- lower ground bar to upper ground bar

Slot 3 (x 23-36, y 90-106)

- F1-5 to X2 1 (E-stop latch)
- K2 5 to X2 2b (seal-in, node X)
- K2 6 to X2 3 (node Y)
- K2 A1 to X2 3b (coil, node Y)
- K2 7 to Mega 43 (motor power sense)
- K2 8 to 0V-S
- door cable: X2 1 to S2 NC1, via the lower left grommet
- door cable: NC1 and S3 (node X) to X2 2
- door cable: S3 (node Y) to X2 3

Notes

Seen from above, front at the left, so the left wall is at the top. Each part sits where sheet 3 puts it; ducts and channels are drawn wider than scale so that every wire shows: measure from sheet 3. Terminals are drawn on the face their wires leave from; read the real order off each part.

Since 2026-10-15 the Pi sits at the operator panel on the 5 V PSU, the Mega runs on the PI's USB and the screen has no supply of its own: F1-1, F1-3, X8 and X2 4-5 are spare (NC2 and the home button are wired at the panel). The ODrive sits in the spin coater's base.

Gauges (issue #205): mains, the PSU to F3 and S, F3 to K2 and FB1, K2's pole jumper, K2 to FB2, S to 0V-P and the 0V-P link: 12 AWG copper. F2-1 to F2-8 with their 0 V returns, and the servo 6 V and Pi 5 V pairs: 14 AWG copper. The F0 rung: 18 AWG UL 1015, 600 V. Everything else: the 20 AWG on hand, copper-clad aluminum (about 3 A), only on fuses of 3 A or less.

PE home runs of the frame, the DIN rails and the TB6600 cases go to the ground bars (sheet 4). 0 V meets PE only at star S, by its one wire to the lower ground bar.

Open: tower common polarity, lid feed, valve coil, servo 6 V rating, J11 logic supply, and the 5 V PSU's current for the Pi 5 (5.1 V, 5 A).

Motor cables: shield to the clamp bar; six leave by the upper left grommet and one by the upper right one: pick which on the bench. Each TB6600: PUL-, DIR- and ENA- jumpered to COM at the driver. The tower cable leaves X7 by the upper right grommet, as sheet 3 has it, so it crosses the deck behind every rail; the upper left grommet, beside X7, would save most of that run.

1 k from 5V to 43 and 100 nF from 40 to GND on the Mega's screw shield.

RMR control panel: upper deck wiring

Sheet 2B (2A: lower deck). Legal 14 x 8.5 in, landscape, 100 %.
Robot-MakingRobot, docs/wiring. Drawn 2026-10-19 (issue #164), revised 2026-10-22.
Status: design, not yet built or tested.
Circuits: sheet 1. Positions: sheet 3. Frame and PE: sheet 4.

RMR control panel: names and terminals

Sheet 2C: what each name on sheets 1, 2A and 2B is and where it sits, and what lands on every terminal. The terminal plan is read from the wire lists that draw sheets 2A and 2B.

Lower deck: 110 VAC and 24 V power

Q0	switch on the IEC inlet: isolates the mains	back wall
X0	mains in: L, L, N, N, PE from the inlet	110 V rail
F0	fuse holder, 1 A T: feeds the K1 rung	110 V rail
K1	110 VAC contactor HC1-63: mains to the 24 V PSU, coil by S1	110 V rail
X1	door 110 V: out to S1 and H1 on the panel	110 V rail
PE	4 ground blocks: inlet PE, PSU FG, door plate, lower bar	110 V rail
PSU	24 V 1200 W supply, lying flat, fan to the right wall	110 VAC area
FB1	12-way fuse module: +24 V always on	fuse rail
FB2	12-way fuse module: +24 V cut by the E-stop (behind K2)	fuse rail
F3	3-pole fuse holder RT18-32, 10 x 38 mm: F3-1 25 A from PSU V+ to K2 (FB2), F3-2 10 A to FB1, F3-3 spare	24 V rail
K2	24 V contactor BMC7-25U: 1-2 and 3-4 in series feed FB2, 5-6 seal in, 7-8 tell Mega pin 43 that motor power is on	24 V rail
star S	the 0 V star at the PSU's -V; the only 0 V to PE bond	24 V rail
0V-P 1, 2	0 V power buses, linked: drivers, ODrive, K2 coil, loads	24 V rail
servo PSU	6 V DC-DC for the gripper and lid servos	beside 24 V rail
5 V PSU	feeds the Pi at the operator panel	24 V row, back
ground bar	lower PE bar on the base plate's PE stud	behind servo PSU

Upper deck: drives, logic, field terminals

X2	E-stop latch: the door cable from S2 and S3	field rail
X4	endstops: a signal and a common for each axis	field rail
X5	servos: 6 V, 0 V and signal for the gripper and lid	field rail
X6	lid (UV array, switch S5) and the argon valve	field rail
X7	status tower: red, amber, green, buzzer, common	field rail
X8	spare (was the screen supply and the home button)	field rail
Mega	Arduino Mega 2560 on a screw shield; runs on the Pi's USB	logic rail
0V-S	signal 0 V bus, 9 linked: driver COMs, Mega GND, endstops	logic rail
0V-S2	second 0V-S group: SSR input returns, S5 return	logic rail
SSR-UV	DC solid-state relay: Mega pin 4 switches the UV array	SSR rail
SSR-valve	DC solid-state relay: Mega pin 42 switches the argon valve	SSR rail
tower R A G	DC SSRs: Pi GPIO 5, 6, 13 switch the red, amber, green lamps	SSR rail
buzzer	DC SSR: Pi GPIO 26 switches the tower's buzzer	SSR rail
X ... C	TB6600 drivers: X, Y1, Y2 (follows Y), Z, A (filter feed), B (syringe height), C (syringe)	drivers rail
shield clamp	grips the motor cables' shields, bonded to PE	past the drivers
ground bar	upper PE bar on the upper plate's PE stud	behind back run

Outside the box (sheet 1)

S1, H1	panel knob that switches K1 on; H1 its lamp, lit while K1 is in	operator panel
S2	E-stop mushroom: NC1 drops K2, NC2 tells the Pi (GPIO 20)	operator panel
S3	Reset push button (NO): pulls K2 in after an E-stop	operator panel
S4	lid switch in series with the UV array: closed = lid shut	lid
S5	lid switch to Mega pin 39 (UV_LID_INTERLOCK, off for now)	lid
S6	Start / Resume push button (NO, green): Pi GPIO 16, Run All or resume a hold	operator panel
S7	Stop push button (NO, red): Pi GPIO 25, M410 and a soft stop of the run	operator panel
D1	1N4007 across the argon valve's coil, band to +	at the valve
Pi	Raspberry Pi 5 with the touch screen, runs rmr_bridge.py	operator panel
ODrive	ODrive S1 spin coater drive; J11 is its UART	spin coater base
X, Y (nodes)	E-stop latch nodes: X after NC1, Y after S3 (sheet 1, rung B)	

Terminal plan

X2 (upper deck)

No.	label	in the box (by)	field side (by)
1	1	door cable, LL (slot 3); F1-5 (slot 3)	+24 V to S2 NC1
2	2	door cable, LL (slot 3)	NC1 out, S3 in (node X)
3	2b	K2 5 (slot 3)	-
4	3	door cable, LL (slot 3); K2 6 (slot 3)	S3 out (node Y)
5	3b	K2 A1 (slot 3)	-
6	4	-	spare: NC2 is wired at the panel
7	5	-	spare

X4 (upper deck)

1	3	Mega 3	endstop X signal (UL)
2	14	Mega 14	endstop Y signal (UL)
3	40	Mega 40	endstop Z signal (UL)
4	15	Mega 15	endstop A signal (UL)
5	23	Mega 23	endstop B signal (UL)
6	A9	Mega A9	endstop C signal (UL)
7	C	0V-S	endstop X common (UL)
8	C	bridged to 7	endstop Y common (UL)
9	C	bridged to 7	endstop Z common (UL)
10	C	bridged to 7	endstop A common (UL)
11	C	bridged to 7	endstop B common (UL)
12	C	bridged to 7	endstop C common (UL)

X5 (upper deck)

1	6V	servo PSU OUT+ (slot 2)	gripper servo +6 V (UL)
2	6V	bridged to 1	lid servo +6 V (UL)
3	0V	servo PSU OUT- (slot 2)	gripper servo 0 V (UL)
4	0V	bridged to 3	lid servo 0 V (UL)
5	P0	Mega 5	gripper servo signal (UL)
6	P1	Mega 6	lid servo signal (UL)

X6 (upper deck)

1	UV+	SSR-UV out - (slot 2)	S4, then the UV array + (UL)
2	UV-	0V-P 2 (slot 2)	UV array - (UL)
3	S5	Mega 39	lid switch S5 (UL)
4	S5c	0V-S2	S5's other side (UL)
5	V+	SSR-valve out -	argon valve + (D1 across) (UL)
6	V-	0V-P 2 (slot 2)	argon valve - (UL)

X7 (upper deck)

1	R	tower R out -	tower red (UR)
2	A	tower A out -	tower amber (UR)
3	G	tower G out -	tower green (UR)
4	Bz	buzzer out -	tower buzzer (UR)
5	C	0V-P 2 (slot 2)	tower common (UR)

X8 (upper deck)

1	-	-	-
2	-	-	-
3	-	-	-
4	-	-	-

0V-S (upper deck)

1	X	K2 8 (slot 3); driver X COM	-
2	Y1	Mega GND; driver Y1 COM	-
3	Y2	X4 7; driver Y2 COM	-
4	Z	driver Z COM	-
5	A	driver A COM	-
6	B	driver B COM	-
7	C	driver C COM	-
8	J	bridged to 1	ODrive J11 GND (UF)
9	lk	0V-S2	-

0V-S2 (upper deck)

1	1	SSR-UV in -	-
2	2	SSR-valve in -	-
3	3	X6 4	-
4	lk	0V-S	-

X0 (lower deck)

1	L	K1 1	L from the inlet (inlet)
2	L	F0 in	-
3	N	K1 3	N from the inlet (inlet)
4	N	K1 A2; X1 3	-
5	PE	PE 1	PE from the inlet (inlet)

X1 (lower deck)

1	1	F0 out	to S1 (gland)
2	2	K1 A1	from S1: K1 coil, H1 (gland)
3	3	X0 4	N to H1 (gland)

PE (lower deck)

1	PE	X0 5	-
2	PE	PSU FG	-
3	PE	-	door plate PE (gland)
4	PE	ground bar	-

Buses on the lower deck

star S	0V-P 1; PSU -V; 5 V PSU IN-; ground bar; servo PSU IN-
0V-P 1	driver A 0V (slot 1); driver B 0V (slot 1); driver C 0V (slot 1); star S; K2 A2; 0V-P 2; driver X 0V (slot 1); driver Y1 0V (slot 1); driver Y2 0V (slot 1); driver Z 0V (slot 1)
0V-P 2	0V-P 1; ODrive DC- (LL); X7 5 (slot 2); X6 2 (slot 2); X6 6 (slot 2)

Fuses

FB1	F1-1 spare, F1-2 5 V PSU (3 A), F1-3 spare, F1-4 tower (3 A), F1-5 latch (3 A), F1-6 spare, F1-7 spare, F1-8 spare, F1-9 spare, F1-10 spare, F1-11 spare, F1-12 spare
FB2	F2-1 X (5 A), F2-2 Y1 (5 A), F2-3 Y2 (5 A), F2-4 Z (5 A), F2-5 A (5 A), F2-6 B (5 A), F2-7 C (5 A), F2-8 ODrive (10 A), F2-9 servo (3 A), F2-10 UV (2 A), F2-11 valve (2 A), F2-12 spare
F0	1 A T, 5 x 20 mm: the K1 rung (S1, K1 coil, H1)
F3	10 x 38 mm, DC rated: F3-1 25 A, PSU V+ to K2 (FB2); F3-2 10 A, PSU V+ to FB1; F3-3 spare

Cable entries and slots

UL	upper left grommet: field cables X4, X5, X6
ULM	upper left grommet 2: 6 motor cables, gantry PE
UF	upper front grommet: Mega USB, J11, Pi GPIO
UR	upper right grommet: X7 tower, 1 motor cable
LL	lower left grommet: ODrive 24 V, Pi 5 V, door cable
inlet	IEC inlet with switch Q0, back panel
gland	M20 gland, back panel: door 110 V cable
slot 1	upper plate: driver +V and 0 V
slot 2	upper plate: SSR feeds, returns, servo 6 V, PE link
slot 3	upper plate: E-stop latch, pin 43, door cable

Abbreviations

PE	protective earth
FG	the PSU's frame ground (earth) terminal
NC, NO	normally closed, normally open contact
SSR	solid-state relay
COM	common
PUL, DIR, ENA	a TB6600's step, direction, enable inputs
F1-n, F2-n, F3-n	fuse n of FB1, of FB2, of F3
0V-P, 0V-S	0 V power, 0 V signal
GPIO	a Pi input/output pin (BCM numbers)
J11	the ODrive's isolated UART connector
RX2, TX2	the Mega's Serial2 pins (17, 16)
USLKG 5	DIN ground terminal block
UK 2.5 B	DIN feed-through terminal block
TS35	35 mm top-hat DIN rail
SJOOW	oil-resistant portable cord, 300 V
AWG	American wire gauge

RMR control panel: names and terminals

Sheet 2C (2A, 2B: the decks; 2D: the panel). Legal 14 x 8.5 in, 100 %.
Robot-MakingRobot, docs/wiring. Drawn 2026-10-20 (issue #171).
Status: design, not yet built or tested.
Circuits: sheet 1. Positions: sheet 3. Wiring: sheets 2A, 2B.

RMR control panel: operator panel

Sheet 2D: how the operator panel might look from the front, at 1:1.75 with assumed sizes, and what each control wires to. The dashed outline is the panel as it may grow.



Behind the panel

control	contact	lands on	cable (by)
S1 power knob	NO, 110 V	X1 1 in, X1 2 out	110 V door cable (M20 gland)
H1 lamp, in S1	110 V	X1 2 and X1 3 (N)	110 V door cable
panel PE		PE 3	110 V door cable
S2 E-stop	NC1	X2 1 (+24 V), X2 2 (node X)	E-stop loop (LL, slot 3)
S2 E-stop	NC2	Pi GPIO 20 and GND, 1 k to 3V3	at the panel
S3 Reset	NO	X2 2 (node X) to X2 3 (node Y)	E-stop loop (LL, slot 3)
S6 Start	NO	Pi GPIO 16 and GND	at the panel
S7 Stop	NO	Pi GPIO 25 and GND	at the panel
touch screen	HDMI, USB	Pi 5	at the panel
Pi 5	5 V in	5 V PSU (F1-2)	5 V feed (LL)
Pi 5	USB	Mega	USB cable (UF)
Pi 5	GPIO 5, 6, 13, 26	tower SSRs (GND to their inputs -)	tower inputs (UF)

Front plate (measured 2026-11-11)

S1, S2, S3, S6, S7: a 22.3 mm hole each; their collars measure 21.8 mm. Cut one test hole first: 22.3 mm is the usual 22 mm device hole, not read off these parts' listings.
Plate 5 to 6 mm thick at most, less any legend plate under a collar (S2's yellow plate): the collar nuts reach no further.
S2's base clamps with two angled screws whose round tips bear on the plate's back: keep the back flat and bare round its hole.
Screen: VESA 75 x 75 on its back. Window, bracket, size: not designed.

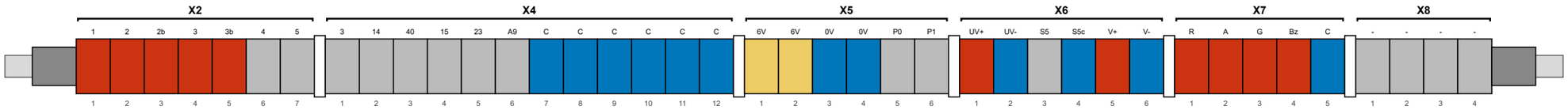
RMR control panel: operator panel

Sheet 2D (2A, 2B: the decks; 2C: names). Legal 14 x 8.5 in, 100 %.
Robot-MakingRobot, docs/wiring. Drawn 2026-10-20 (#171), rev. 2026-11-11 (#218).
Status: concept; housing and sizes not designed yet.
Circuits: sheet 1. Terminals: sheet 2C.

RMR control panel: terminal block colors

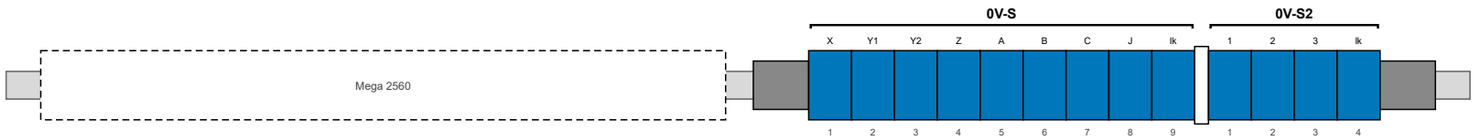
Sheet 2E: the color of every terminal block, read from the wire lists that draw sheets 2A and 2B, so it follows the terminal plan. Each rail at 1:1 along its length, front of the box at the left.

Field rail, upper deck: 284 mm



Grey: end clamp
White: end cover
Dashed: not a terminal block
Blocks 6.2 mm wide, printed at 100 %

Logic rail, upper deck: 211.6 mm



110 V rail, lower deck: 154.5 mm



A block takes the color of the wire that lands on it

- 110 VAC line
- 110 VAC neutral (the kits have no white), signals and spares
- +24 V, always on and behind K2
- every 0 V: 0V-P, 0V-S and the supplies' own (blue wire, white bands)
- +6 V servo supply: its wire is red, the block says it is not 24 V
- PE: grounding blocks, which bond to their rail

Never use the kits' plain green blocks: they look like PE and do not bond to the rail. What lands on each terminal: sheet 2C.

Blocks needed

grey	18	the kits and a 100-pack
blue	25	the kits and a 100-pack
red	11	the kits
black	4	the kits
yellow	2	the kits
green-yellow	14	USLKG 5 or the kits' grounding blocks, 9 of them one per other DIN rail

The two kits hold 120 blocks in six colors; the listing does not give the split, but its photos show 10 of each per kit, 20 in all. The blue and grey packs in the BoM (issue #209) are the same GUETNEU block and cover those two; count the other colors before wiring. The per-rail blocks are not placed on sheets 2A, 2B and 3 yet (ADR-0014).

Block by block

Block	No.	Color	Wire class
X2	1	red	+24 V always on (FB1)
	2-5	red	+24 V behind K2 (FB2)
	6-7	grey	spare
X4	1-6	grey	signal
	7-12	blue	0V-S signal 0 V
X5	1-2	yellow	+6 V servo, +5 V Pi
	3-4	blue	0 V of the servo and Pi supplies
	5-6	grey	signal
X6	1	red	+24 V behind K2 (FB2)
	2	blue	0 V power (0V-P)
	3	grey	signal
	4	blue	0V-S signal 0 V
	5	red	+24 V behind K2 (FB2)
	6	blue	0 V power (0V-P)
X7	1-4	red	+24 V always on (FB1)
	5	blue	0 V power (0V-P)
X8	1-4	grey	spare
0V-S	1-9	blue	0V-S signal 0 V
0V-S2	1-4	blue	0V-S signal 0 V
X0	1-2	black	110 VAC line
	3-4	grey	110 VAC neutral
	5	green-yellow	PE
X1	1-2	black	110 VAC line
	3	grey	110 VAC neutral
PE	1-4	green-yellow	PE

RMR control panel: terminal block colors

Sheet 2E (2A, 2B: the decks; 2C: terminals). Legal 14 x 8.5 in, 100 %.
Robot-MakingRobot, docs/wiring. Drawn 2026-10-22 (issue #207).
Status: design, not yet built.
Colors from the wire lists of sheets 2A and 2B.